

HW 6 - Zeshui Song

Note:

Source Free RC and RL Circuit Response

RC and RL circuits with a single energy storage element (a single capacitor or a single inductor):

1. Determine the time constant using a Thevenin equivalent circuit.

$$\tau_c = R_{Th}C, \quad \tau_L = \frac{L}{R_{Th}}$$

2. Find the initial conditions (v_0 for RC circuits and i_0 for RL circuits).

3. Solution:

$$v(t) = v_0 e^{-\frac{t}{\tau}}, \quad i(t) = i_0 e^{-\frac{t}{\tau}}$$

Driven RC Circuit Step Response

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} and C_{eq} , and the time constant $\tau_{eq} = R_{eq}C_{eq}$.
2. Determine the initial condition $v(0^+)$ or $i(0^+)$ [recall that the capacitor voltage is continuous: $v_c(0^-) = v_c(0^+)$]
3. Determine the steady state condition $v(\infty)$ or $i(\infty)$.
4. The final response is given by:

$$\begin{aligned} v(t) &= v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau_{eq}} \\ i(t) &= i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau_{eq}} \end{aligned}$$

Driven RL Circuit Step Response

1. With all independent sources zeroed out, simplify the circuit to determine R_{eq} and L_{eq} , and the time constant $\tau_{eq} = L_{eq}/R_{eq}$.
2. Determine the initial condition $v(0^+)$ or $i(0^+)$ [recall the requirement that any inductor current $i_L(0^-) = i_L(0^+)$].
3. Determine the final condition $v(\infty)$ or $i(\infty)$.
4. The final response is given by

$$\begin{aligned} v(t) &= v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau_{eq}} \\ i(t) &= i(\infty) + [i(0^+) - i(\infty)]e^{-t/\tau_{eq}} \end{aligned}$$

- 8.2** 2. A source-free RC circuit has $v(0) = 12 \text{ V}$ and $R = 100 \Omega$. (a) Select C such that $v(250 \mu\text{s}) = 5.215 \text{ V}$; (b) compute the energy stored in the capacitor at $t = 0$, $t = 250 \mu\text{s}$, $t = 500 \mu\text{s}$, and $t = 1 \text{ ms}$.

The KCL for a source free RC circuit is given by:

$$C \frac{dv}{dt} + \frac{v}{R} = 0$$

Solving by separating and integrating, we have:

$$\begin{aligned} \int \frac{1}{v} dv &= - \int \frac{1}{RC} dt \\ \ln |v| &= -\frac{t}{RC} + C \\ v(t) &= e^C e^{-\frac{t}{RC}} = C_1 e^{-\frac{t}{RC}} \end{aligned}$$

Plugging in the initial condition $v(0) = v_0$, we solve for C_1 :

$$v(0) = C_1 e^0 = C_1$$

Thus, $C_1 = v_0$, and the solution is:

$$\boxed{v(t) = v_0 e^{-\frac{t}{RC}}}$$

(a) Using the equation for $v(t)$, we plug in the values for v_0 , R , and $v(250 \mu\text{s})$ to solve for C :

$$v(250 \times 10^{-6} \text{ s}) = [5.215 \text{ V}] = [12 \text{ V}] \exp \left(-\frac{[250 \times 10^{-6} \text{ s}]}{[100 \Omega]C} \right)$$

$$\boxed{C = 2.999 \times 10^{-6} \text{ F}}$$

(b) The energy stored in the capacitor is given by:

$$w = \frac{1}{2} C v^2$$

The voltages are:

$$v(0) = 12 \text{ V}$$

$$v(250 \times 10^{-6} \text{ s}) = 5.215 \text{ V}$$

$$v(500 \times 10^{-6} \text{ s}) = 2.265 \text{ V}$$

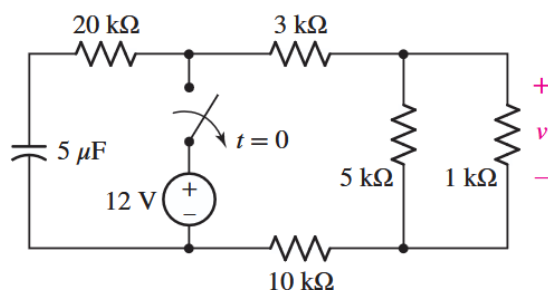
$$v(1 \times 10^{-3} \text{ s}) = 0.427 \text{ V}$$

Thus, the energies are:

$$\begin{aligned} w(0) &= \frac{1}{2} [2.999 \times 10^{-6} \text{ F}] [12 \text{ V}]^2 = 2.159 \times 10^{-4} \text{ J} \\ w(250 \times 10^{-6} \text{ s}) &= \frac{1}{2} [2.999 \times 10^{-6} \text{ F}] [5.215 \text{ V}]^2 = 4.078 \times 10^{-5} \text{ J} \\ w(500 \times 10^{-6} \text{ s}) &= \frac{1}{2} [2.999 \times 10^{-6} \text{ F}] [2.265 \text{ V}]^2 = 7.692 \times 10^{-6} \text{ J} \\ w(1 \times 10^{-3} \text{ s}) &= \frac{1}{2} [2.999 \times 10^{-6} \text{ F}] [0.427 \text{ V}]^2 = 2.734 \times 10^{-7} \text{ J} \end{aligned}$$

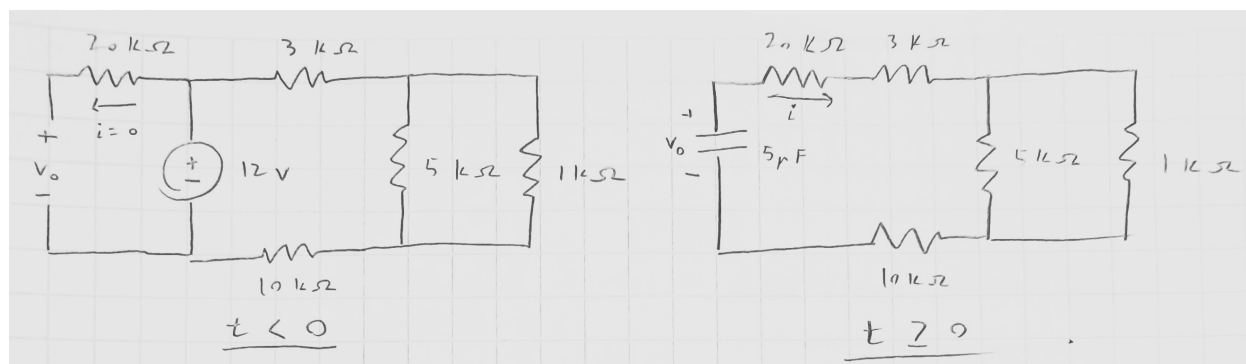
8.8

8. The switch above the 12 V source in the circuit of Fig. 8.54 has been closed since just after the wheel was invented. It is finally thrown open at $t = 0$. (a) Compute the circuit time constant. (b) Obtain an expression for $v(t)$ valid for $t > 0$. (c) Calculate the energy stored in the capacitor 170 ms after the switch is opened.



■ FIGURE 8.54

Redrawing the circuit before and after the switch is closed, we have:



Note: After the switch is flipped, the current will change directions as the capacitor discharges, letting current flow out from its (+) terminal.

Since a capacitor at steady state behaves like an open circuit, there is no voltage drop across the 20 k Ω resistor ($i = 0$), thus the voltage across the capacitor is $v_0 = 12$ V.

(a) The time constant is given by: $\tau = R_{Th}C$. The Thevenin equivalent resistance just the sum of resistances (after opening the switch at $t=0$) since the source is independent:

Combine the 1 k Ω and 5 k Ω resistors in parallel:

$$R_1 = \frac{[1 \cdot 5 \text{ k}\Omega]}{[1 + 5 \text{ k}\Omega]} = 0.833 \text{ k}\Omega$$

Combine everything else in series:

$$R_{Th} = R_1 + [20 \text{ k}\Omega] + [3 \text{ k}\Omega] + [10 \text{ k}\Omega] = 33.833 \text{ k}\Omega$$

Thus, the time constant is:

$$\tau = R_{Th}C = [33.833 \times 10^3 \Omega][5 \times 10^{-6} \text{ F}] = 169.16 \times 10^{-3} \text{ s}$$

(b) Since we know both v_0 and τ , we can solve for the voltage across the capacitor for $t > 0$:

$$v_c = 12 \exp\left(-\frac{t}{169.16 \times 10^{-3}}\right)$$

To find $v(t)$, we can use voltage division for the voltage across the 5 k Ω - 1 k Ω parallel combination:

$$v(t) = \frac{R_1}{R_1 + [20 \text{ k}\Omega] + [3 \text{ k}\Omega] + [10 \text{ k}\Omega]} v_c = 0.295 \exp\left(-\frac{t}{169.16 \times 10^{-3}}\right)$$

$$v(t) = 0.295 \exp\left(-\frac{t}{169.16 \times 10^{-3}}\right) [\text{V}]$$

(c) The energy stored in the capacitor is given by:

$$w = \frac{1}{2} C v_c^2$$

$$v_c(170 \times 10^{-3} \text{ s}) = 12 \exp\left(-\frac{170 \times 10^{-3}}{169.16 \times 10^{-3}}\right) = 4.392 \text{ V}$$

$$w(170 \times 10^{-3} \text{ s}) = \frac{1}{2} [5 \times 10^{-6} \text{ F}] [4.392 \text{ V}]^2 = 4.823 \times 10^{-5} \text{ J}$$

$$w(170 \times 10^{-3} \text{ s}) = 4.823 \times 10^{-5} \text{ J}$$

- 8.20** 20. Referring to the circuit shown in Fig. 8.11, select values for both elements such that $L/R = 1$, $i(0) = -5$ A, and $v_R(0) = 10$ V. (a) Calculate $v_R(t)$ at $t = 0, 1, 2, 3, 4$, and 5 s; (b) compute the power dissipated in the resistor at $t = 0, 1$ s, and 5 s. (c) At $t = 5$ s, what is the percentage of the initial energy still stored in the inductor?

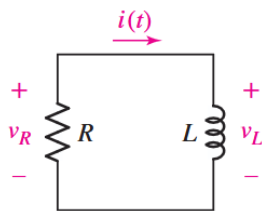


FIGURE 8.11 A series RL circuit for which $i(t)$ is to be determined, subject to the initial condition that $i(0) = i_0$.

Given:

- $\tau = \frac{L}{R} = 1$
- $i(0) = -5$ A
- $v_R(0) = 10$ V

The current within the RL circuit is given by:

$$i(t) = i_0 \exp\left(-\frac{t}{\tau}\right) = -5 \exp(-t)$$

The resistance is then:

$$R = -\frac{v_R(0)}{i(0)} = -\frac{[10 \text{ V}]}{[-5 \text{ A}]} = 2 \Omega$$

Note: Here, the current *doesn't* follow the passive sign convention since it terms the (-) terminal of the resistor instead of the (+) terminal. Thus, Ohms law becomes $v_R = -iR$.

The inductance is then:

$$L = R \implies L = 2 \text{ H}$$

(a) The voltage across the resistance is then:

$$v_R(t) = i(t)R = -5 \exp(-t) \cdot 2 = -10 \exp(-t)$$

$v_R(0) = -10 \exp(0) = -10 \text{ V}$
$v_R(1) = -10 \exp(-1) = -3.678 \text{ V}$
$v_R(2) = -10 \exp(-2) = -1.353 \text{ V}$
$v_R(3) = -10 \exp(-3) = -0.497 \text{ V}$
$v_R(4) = -10 \exp(-4) = -0.183 \text{ V}$
$v_R(5) = -10 \exp(-5) = -0.067 \text{ V}$

(b) Power dissipated by the resistor is given by:

$$P_R = \frac{V_R^2}{R}$$

$$\begin{aligned}
 P(0) &= \frac{[10 \text{ V}]^2}{2 \Omega} = 50 \text{ W} \\
 P(1) &= \frac{[3.678 \text{ V}]^2}{2 \Omega} = 6.763 \text{ W} \\
 P(5) &= \frac{[0.067 \text{ V}]^2}{2 \Omega} = 0.002 \text{ W}
 \end{aligned}$$

(c) The energy stored in the inductor is given by:

$$w = \frac{1}{2} L i^2$$

Thus,

$$w(0) = \frac{1}{2} [2 \text{ H}] [-5 \text{ A}]^2 = 25 \text{ J}$$

$$w(5) = \frac{1}{2} [2 \text{ H}] [-5 \exp(-5)]^2 = 0.00113 \text{ J}$$

Percentage of energy left in the inductor at $t = 5 \text{ s}$ is given by:

$$\frac{w(5)}{w(0)} \times 100 = \frac{[0.00113 \text{ J}]}{[25 \text{ J}]} \times 100 = \boxed{0.0045\%}$$

- 8.23** 23. The switch in Fig. 8.57 has been closed since Catfish Hunter last pitched for the New York Yankees. Calculate the voltage labeled v as well as the energy stored in the inductor at (a) the instant just prior to the switch being thrown open; (b) the instant just after the switch is opened; (c) $t = 8 \mu\text{s}$; (d) $t = 80 \mu\text{s}$.

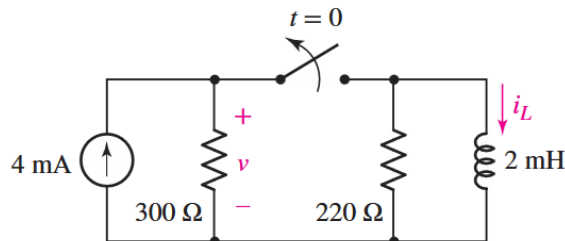
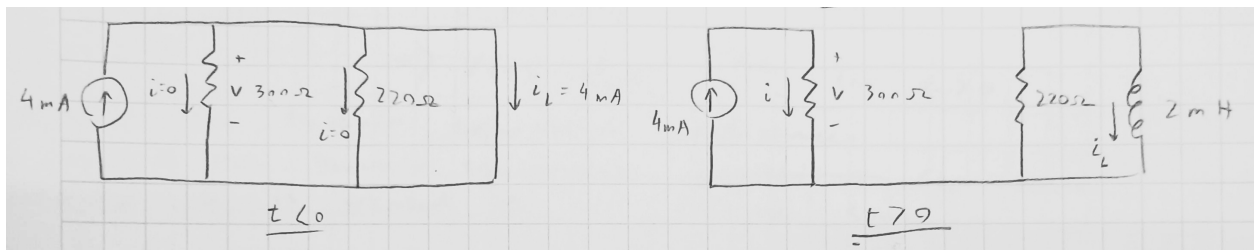


FIGURE 8.57

At steady state, the inductor behaves like a short circuit. Since the inductor is in parallel with both the 300Ω and 220Ω resistors, there will be ZERO current through either resistor because current flows through the path of least resistance.

$$\Rightarrow i_L(0) = 4 \text{ mA}$$



Note: Inductors resist changes in current, so after being disconnected, it will continue to supply current in the same direction as before.

(a) Just before the switch is open:

$$v = 0$$

The energy stored in the inductor is given by:

$$w = \frac{1}{2}Li_L^2 = \frac{1}{2}[2 \times 10^{-3} \text{ H}][4 \times 10^{-3} \text{ A}]^2 = \boxed{16 \times 10^{-9} \text{ J}}$$

(b) Just after the switch is open:

$$v = iR = [4 \times 10^{-3} \text{ A}] \cdot [300 \Omega] = 1.2 \text{ V}$$

Time constant is given by:

$$\tau = \frac{L}{R} = \frac{[2 \times 10^{-3} \text{ H}]}{[220 \Omega]} = 9.09 \times 10^{-6} \text{ s}$$

Thus,

$$i_L(t) = i_L(0) \exp\left(-\frac{t}{\tau}\right) = (4 \times 10^{-3}) \exp\left(-\frac{t}{9.09 \times 10^{-6}}\right)$$

Since the current hasn't changed yet ($t=0$), the energy stored in the inductor is still:

$$w = \boxed{16 \times 10^{-9} \text{ J}}$$

(c) & (d) Using the formula for current:

$$i_L(8 \times 10^{-6}) = (4 \times 10^{-3}) \exp\left(-\frac{8 \times 10^{-6}}{9.09 \times 10^{-6}}\right) = 1.658 \times 10^{-3} \text{ A}$$

$$i_L(80 \times 10^{-6}) = (4 \times 10^{-3}) \exp\left(-\frac{80 \times 10^{-6}}{9.09 \times 10^{-6}}\right) = 0.602 \times 10^{-6} \text{ A}$$

Thus, the energies are:

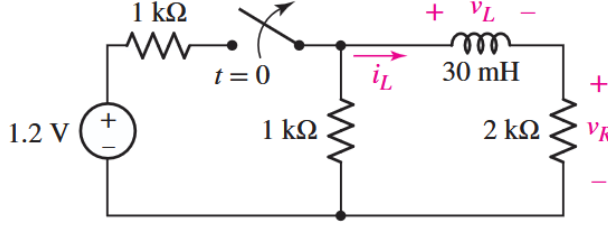
$$w(8 \times 10^{-6}) = \frac{1}{2}Li_L^2 = \frac{1}{2}[2 \times 10^{-3} \text{ H}][1.658 \times 10^{-3} \text{ A}]^2 = \boxed{2.748 \times 10^{-9} \text{ J}}$$

$$w(80 \times 10^{-6}) = \frac{1}{2}Li_L^2 = \frac{1}{2}[2 \times 10^{-3} \text{ H}][0.602 \times 10^{-6} \text{ A}]^2 = \boxed{3.624 \times 10^{-16} \text{ J}}$$

The voltage is constant for the left hand side:

$$v = 1.2 \text{ V}$$

- 8.28** 28. The switch shown in Fig. 8.62 had been closed for 6 years prior to being flipped open at $t = 0$. Determine i_L , v_L , and v_R at t equal to (a) 0^- ; (b) 0^+ ; (c) $1 \mu\text{s}$; (d) $10 \mu\text{s}$.



■ **FIGURE 8.62**

(a) $t = 0^-$ (Just before the switch is opened): At steady state, the inductor behaves like a short circuit and $di/dt = 0$. Thus,

$$\boxed{V_L = 0}$$

V_R is the same as the voltage across the $1 \text{ k}\Omega$ - $2 \text{ k}\Omega$ parallel combination. Thus, by voltage division:

$$R_p = \frac{[1 \cdot 2 \text{ k}\Omega]}{[1 + 2 \text{ k}\Omega]} = 0.666 \text{ k}\Omega$$

$$V_R = \frac{[0.666 \text{ k}\Omega]}{[1 + 0.666 \text{ k}\Omega]} \cdot [1.2 \text{ V}] = 0.4801 \text{ V}$$

$$\boxed{V_R = 0.4801 \text{ V}}$$

The current i_L is equal to the current through the $2 \text{ k}\Omega$ resistor (series connection):

$$i_L = \frac{V_R}{2 \text{ k}\Omega} = \boxed{0.2401 \text{ mA}}$$

(b) $t = 0^+$ (Just after the switch is opened): Not enough time has passed for the current to change, so i_L is still:

$$\boxed{i_L = 0.2401 \text{ mA}}$$

The voltage across the inductor can be found using KVL:

$$-V_L - [2 \text{ k}\Omega] \cdot i_L - [1 \text{ k}\Omega] \cdot i_L = 0 \implies V_L = -0.7202 \text{ V}$$

$$\boxed{V_L = -0.7202 \text{ V}}$$

Also,

$$\boxed{V_R = [2 \text{ k}\Omega] \cdot i_L = 0.4801 \text{ V}}$$

(c) The time constant is given by:

$$\tau = \frac{L}{R} = \frac{[30 \times 10^{-3} \text{ H}]}{[3 \times 10^3 \Omega]} = 1 \times 10^{-5} \text{ s}$$

Thus, the current is given by:

$$i_L(t) = i_L(0) \exp\left(-\frac{t}{\tau}\right) = (0.2401 \times 10^{-3}) \exp\left(-\frac{t}{1 \times 10^{-5}}\right)$$

$$i_L(10^{-6}) = 0.2172 \times 10^{-3} \text{ A}$$

Voltage is given by:

$$\Rightarrow V_L = L \frac{di_L}{dt} = [30 \times 10^{-3} \text{ H}] \cdot \frac{d}{dt} \left[(0.2401 \times 10^{-3}) \exp \left(-\frac{t}{1 \times 10^{-5}} \right) \right]$$

$$V_L = -0.7203 \exp \left(-\frac{t}{10^{-5}} \right)$$

$$V_L(10^{-6}) = -0.6519 \text{ V}$$

Thus, the voltage across the resistor is:

$$V_R(10^{-6}) = [2 \text{ k}\Omega] \cdot i_L(10^{-6}) = 0.4344 \text{ V}$$

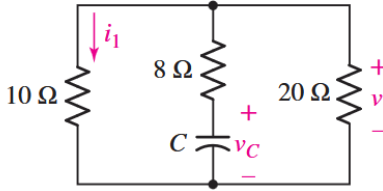
(d)

$$i_L(10 \times 10^{-6}) = 0.8832 \times 10^{-4} \text{ A}$$

$$V_L(10 \times 10^{-6}) = -0.2649 \text{ V}$$

$$V_R(10 \times 10^{-6}) = 0.1766 \text{ V}$$

- 8.35** 35. A quick measurement determines that the capacitor voltage v_C in the circuit of Fig. 8.67 is 2.5 V at $t = 0^-$. (a) Determine $v_C(0^+)$, $i_1(0^+)$, and $v(0^+)$. (b) Select a value of C so that the circuit time constant is equal to 14 s.



■ **FIGURE 8.67**

(a) The capacitor voltage is continuous, so

$$V_C(0^-) = V_C(0^+) = 2.5 \text{ V}$$

The voltage V is the same as the voltage across the 10Ω and 20Ω resistors in parallel. Thus, by voltage division:

$$R_p = \frac{[10 \cdot 20 \Omega]}{[10 + 20 \Omega]} = 6.667 \Omega$$

$$V(0^+) = \frac{[6.667 \Omega]}{[6.667 + 8 \Omega]} \cdot [2.5 \text{ V}] = \boxed{1.136 \text{ V}}$$

Using Ohm's law:

$$i_1(0^+) = \frac{V_{10\Omega}}{[10 \Omega]} = \frac{1.136 \text{ V}}{10 \Omega} = \boxed{0.1136 \text{ A}}$$

(b) The time constant is given by:

$$\tau = R_{Th}C$$

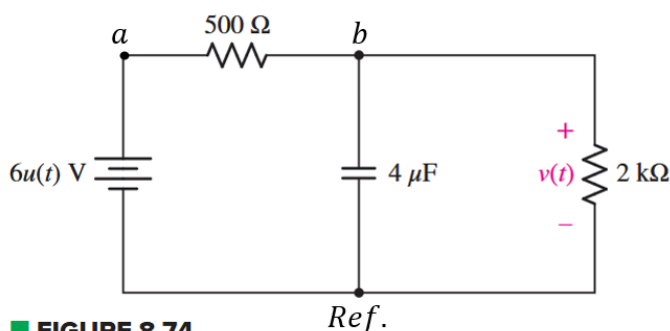
The Thevenin equivalent resistance is the sum of resistances:

$$R_{Th} = R_p + [8 \Omega] = 14.667 \Omega$$

Solving for C :

$$C = \frac{\tau}{R_{Th}} = \frac{[14 \text{ s}]}{[14.667 \Omega]} = \boxed{0.9545 \text{ F}}$$

- 8.46** 46. With reference to the circuit depicted in Fig. 8.74, compute $v(t)$ for (a) $t = 0^-$; (b) $t = 0^+$; (c) $t = 2 \text{ ms}$; (d) $t = 5 \text{ ms}$.



■ FIGURE 8.74

The voltage input is a unit step function, recall that:

$$u(t - a) = \begin{cases} 0 & t < a \\ 1 & t \geq a \end{cases}$$

Thus, at both $t = 0^-$ and $t = 0^+$, the voltage across the capacitor is zero since it starts uncharged and voltage across a capacitor cannot change instantaneously (it is continuous). Thus, it acts like a short circuit.

(a) & (b) Since the capacitor is like a short circuit, for both $t = 0^-$ and $t = 0^+$, the voltage across the $2 \text{ k}\Omega$ resistor in parallel with the capacitor is zero.

$$\boxed{V(0^-) = V(0^+) = 0}$$

(c) There are two methods to solve for $v(t)$ for $t > 0$:

Method 1. using Nodal analysis (KCL) for $t > 0$ and solving the resulting differential equation:

Node a:

$$v_a = 6u(t) \text{ V}$$

Node b:

$$\frac{v_a - v_b}{[500 \Omega]} = C \frac{dv_c}{dt} + \frac{v_b}{[2 \times 10^3 \Omega]}$$

$$\frac{6u(t) - v}{[500 \Omega]} = [4 \times 10^{-6} \text{ F}] \frac{dv_c}{dt} + \frac{v}{[2 \times 10^3 \Omega]}$$

Since the capacitor is in parallel with the $2 \text{ k}\Omega$ resistor, $v_c = v_b = v$.

$$\frac{6u(t) - v}{[500 \Omega]} = [4 \times 10^{-6} \text{ F}] \frac{dv}{dt} + \frac{v}{[2 \times 10^3 \Omega]} \implies 24u(t) = 0.008 \frac{dv}{dt} + 5v$$

Taking the laplace transform:

$$\frac{24u(t)}{s} = 0.008sV(s) + 5V(s) \implies \frac{3000u(t)}{s} = sV(s) + 625V(s)$$

$$V(s) = \frac{3000u(t)}{s(s + 625)} = \frac{A}{s} + \frac{B}{s + 625} \quad (\text{Partial fraction})$$

Solving for A and B using the limit method:

$$A = \lim_{s \rightarrow 0} s \left(\frac{3000u(t)}{s(s + 625)} \right) = 4.8u(t)$$

$$B = \lim_{s \rightarrow -625} (s + 625) \left(\frac{3000u(t)}{s(s + 625)} \right) = -4.8u(t)$$

$$V(s) = \frac{4.8u(t)}{s} - \frac{4.8u(t)}{s + 625}$$

Taking the inverse laplace transform:

$$v(t) = 4.8u(t) - 4.8u(t)e^{-625t} \implies \boxed{v(t) = 4.8u(t)(1 - e^{-625t}) [\text{V}]}$$

Now we can just plug in the values for t to find the voltage:

$$\boxed{v(2 \times 10^{-3}) = 3.424 \text{ V}}$$

(d)

$$\boxed{v(5 \times 10^{-3}) = 4.589 \text{ V}}$$

Method 2: Following the textbook's method for solving driven RC circuits:

Time constant:

$$\tau = R_{\text{Th}}C$$

The Thevenin equivalent resistance is the sum of resistances:

$$R_{\text{Th}} = \frac{[500 \cdot 2 \times 10^3 \Omega]}{[500 + 2 \times 10^3 \Omega]} = 400 \Omega$$

$$\tau = [400 \Omega] \cdot [4 \times 10^{-6} \text{ F}] = 1.6 \times 10^{-3} \text{ s}$$

Initial condition: the capacitor starts uncharged, so

$$v(0^-) = v(0^+) = 0$$

Steady state condition: at $t \rightarrow \infty$, the capacitor behaves like an open circuit, so the voltage across the $2 \text{ k}\Omega$ resistor is the same as the voltage across the capacitor. By voltage division:

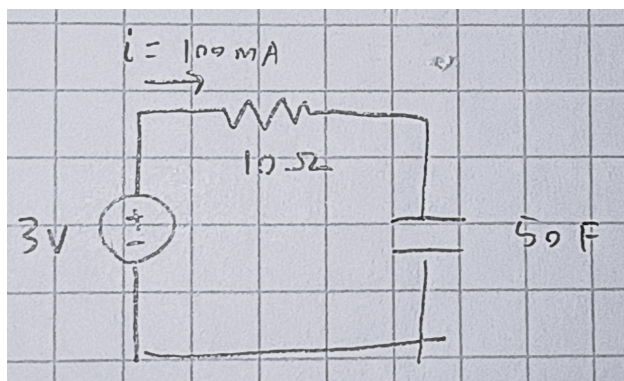
$$v(\infty) = \frac{[2 \times 10^3 \Omega]}{[500 + 2 \times 10^3 \Omega]} \cdot [6 \text{ V}] = 4.8 \text{ V}$$

By the textbook's formula for driven RC circuits:

$$v(t) = v(\infty) + [v(0^+) - v(\infty)]e^{-t/\tau} = 4.8 - 4.8e^{-625t} [\text{V}]$$

This is the same result as Method 1.

- 8.50** 50. You build a portable solar charging circuit consisting of a supercapacitor and a solar cell that provides 100 mA and 3 V . If the series resistance is 10Ω and the supercapacitor is 50 F , determine (a) the total energy storage that can be achieved by the capacitor, and (b) the time it takes to charge to 95 percent of full capacity.



- (a) The energy stored in the capacitor is given by:

$$w = \frac{1}{2} C v^2$$

At steady state when the capacitor is fully charged, there is no voltage drop across the 10Ω resistor, so the voltage across the capacitor is equal to 3 V .

$$w = \frac{1}{2} [50 \text{ F}] [3 \text{ V}]^2 = \boxed{225 \text{ J}}$$

- (b) First we need to find the voltage across the capacitor over time. KCL at the node between the 10Ω resistor and the capacitor gives us:

$$\frac{3 - v}{[10 \Omega]} = C \frac{dv}{dt} \implies 0.3 = 0.1v + 50 \frac{dv}{dt}$$

Taking the laplace transform:

$$\frac{0.3}{s} = 0.1V(s) + 50sV(s)$$

$$\Rightarrow V(s) = \frac{0.3}{s(0.1 + 50s)} = \frac{0.006}{s(0.002 + s)} = \frac{A}{s} + \frac{B}{s + 0.002} \quad (\text{Partial fraction})$$

Solving for A and B using the limit method:

$$A = \lim_{s \rightarrow 0} s \left(\frac{0.006}{s(0.002 + s)} \right) = 3$$

$$B = \lim_{s \rightarrow -0.002} (s + 0.002) \left(\frac{0.006}{s(0.002 + s)} \right) = -3$$

$$V(s) = \frac{3}{s} - \frac{3}{s + 0.002}$$

Taking the inverse laplace transform:

$$v(t) = 3 - 3e^{-0.002t} \text{ [V]}$$

Plug in $v(t)$ to find time to reach 95% of the energy stored at steady state:

$$0.95 \cdot 225 = \frac{1}{2} [50 \text{ F}] [v(t)]^2 \Rightarrow 8.55 = (3 - 3e^{-0.002t})^2$$

$$\Rightarrow t = 1838.06 \text{ s}$$

- 8.54** 54. The switch in the circuit of Fig. 8.80, often called a *make-before-break* switch (since during switching it briefly makes contact with both parts of the circuit to ensure a smooth electrical transition), moves to position b at $t = 0$ only after being in position a long enough to ensure all initial transients arising from turning on the sources have long since decayed. (a) Determine the power dissipated by the 5Ω resistor at $t = 0^-$. (b) Determine the power dissipated in the 3Ω resistor at $t = 2 \text{ ms}$.

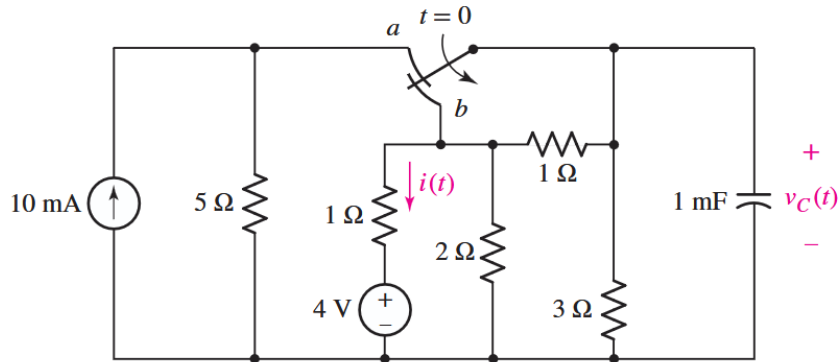
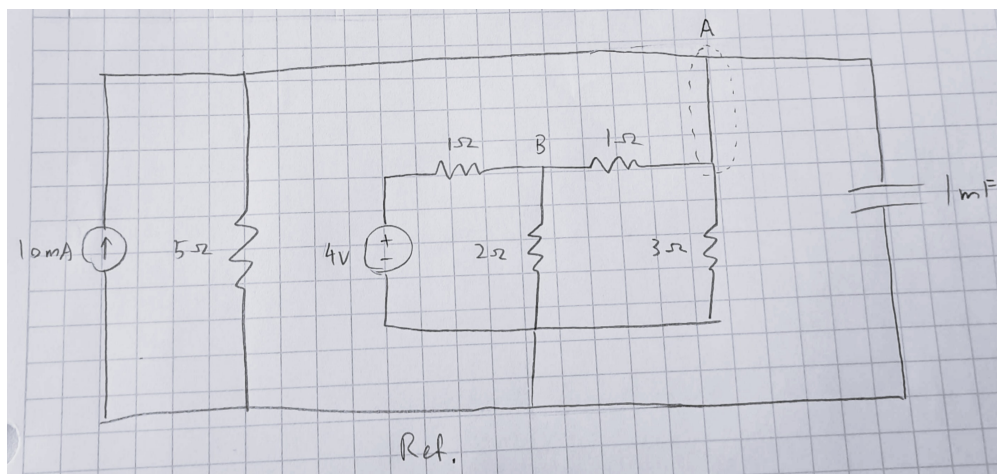


FIGURE 8.80



(a) The voltage across the $5\ \Omega$ resistor is equal to v_A so using nodal analysis at steady state:

Node A:

$$[10\text{ mA}] = \frac{v_A}{[5\ \Omega]} + \frac{v_A}{[3\ \Omega]} + \frac{v_A - v_B}{[1\ \Omega]} \Rightarrow 10 \times 10^{-3} = \frac{23}{15}v_A - v_B$$

Node B:

$$\frac{v_A - v_B}{[1\ \Omega]} = \frac{v_B - 4}{[1\ \Omega]} + \frac{v_B}{[2\ \Omega]} \Rightarrow -4 = v_A - \frac{5}{2}v_B$$

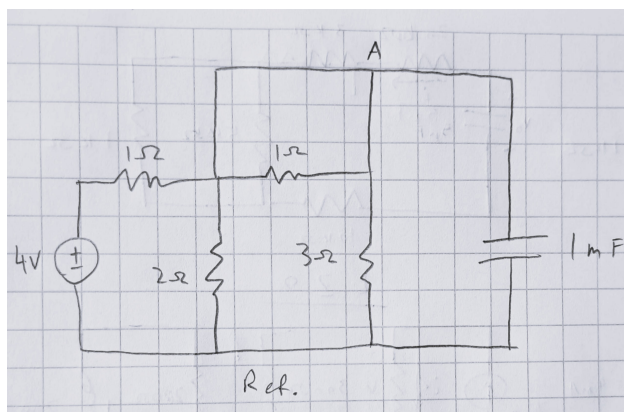
Solving for v_A and v_B :

$$v_A = [1.4206\text{ V}], \quad v_B = [2.1682\text{ V}]$$

Thus, the power dissipated by the $5\ \Omega$ resistor is given by:

$$P = \frac{V^2}{R} = \frac{[1.4206\text{ V}]^2}{[5\ \Omega]} = [0.4036\text{ W}]$$

(b) Redrawing the circuit for the switch set to b for $t > 0$:



Using KCL at node A:

$$-C \frac{dv_A}{dt} = \frac{v_A}{[3\ \Omega]} + \cancel{\frac{v_A - v_A}{[1\ \Omega]}} + \frac{v_A - 4}{[1\ \Omega]} + \frac{v_A}{[2\ \Omega]}$$

Note: The capacitor starts off charged and discharges, so the voltage across the capacitor decreases and $dv_A/dt < 0$. Thus, I have to put a negative sign to keep the current direction consistent.

$$\begin{aligned} \Rightarrow -10^{-3} \frac{dv_A}{dt} &= \frac{11}{6} v_A - 4 \\ \Rightarrow \frac{dv_A}{dt} &= -1833.33 v_A + 4000 \end{aligned}$$

Taking the laplace transform:

$$\begin{aligned} sV_A(s) - v_A(0) &= -1833.33V_A(s) + \frac{4000}{s}, \quad v_A(0) = 1.4206 \text{ V} \\ \Rightarrow V_A(s)(s + 1833.33) &= 1.4206 + \frac{4000}{s} \\ \Rightarrow V_A(s) &= \frac{1.4206}{(s + 1833.33)} + \frac{4000}{s(s + 1833.33)} \end{aligned}$$

Partial fraction decomposition for the second term:

$$\frac{4000}{s(s + 1833.33)} = \frac{A}{s} + \frac{B}{s + 1833.33}$$

Solving for A and B using the limit method:

$$\begin{aligned} A &= \lim_{s \rightarrow 0} s \left(\frac{4000}{s(s + 1833.33)} \right) = 2.18 \\ B &= \lim_{s \rightarrow -1833.33} (s + 1833.33) \left(\frac{4000}{s(s + 1833.33)} \right) = -2.18 \end{aligned}$$

Thus,

$$V_A(s) = \frac{1.4206}{(s + 1833.33)} + \frac{2.18}{s} - \frac{2.18}{s + 1833.33} = \frac{-0.7594}{s + 1833.33} + \frac{2.18}{s}$$

Taking the inverse laplace transform:

$$\boxed{v_A(t) = 2.18 - 0.7594e^{-1833.33t} \text{ [V]}}$$

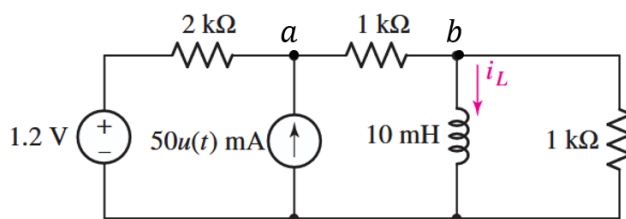
The voltage across the 3Ω resistor is the same as v_A , so the power dissipated by the 3Ω resistor is given by:

$$\begin{aligned} v_A(2 \times 10^{-2}) &= 2.1605 \text{ V} \\ P &= \frac{V^2}{R} = \frac{[2.1605 \text{ V}]^2}{[3 \Omega]} = \boxed{1.556 \text{ W}} \end{aligned}$$

Note: When applying KCL, I usually use the (Current into node) = (Current out of node) convention. However, for inductors and capacitors, it is hard to figure out the direction. Thus, applying *passive sign convention*, we say that ALL currents are exiting the node and NO current is entering the node. This way, the math will naturally work out, and we don't have to worry about the direction of current for inductors and capacitors (one of the terms will be forced negative).

8.61

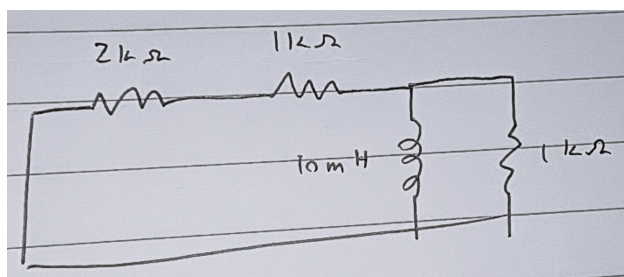
61. The circuit depicted in Fig. 8.86 contains two independent sources, one of which is only active for $t > 0$. (a) Obtain an expression for $i_L(t)$ valid for all t ; (b) calculate $i_L(t)$ at $t = 10 \mu\text{s}$, $20 \mu\text{s}$, and $50 \mu\text{s}$.



■ FIGURE 8.86

Ref.

(a) I will try the textbook method for solving driven circuits here. First turning off all the independent sources (voltage sources become short circuits and current sources become open circuits):



Add $2 \text{ k}\Omega$ and $1 \text{ k}\Omega$ in series:

$$R = 2 + 1 = 3 \text{ k}\Omega$$

Add the $3 \text{ k}\Omega$ and $1 \text{ k}\Omega$ in parallel:

$$R_{\text{eq}} = \frac{3 \cdot 1}{3 + 1} = 0.75 \text{ k}\Omega$$

Thus the time constant is given by:

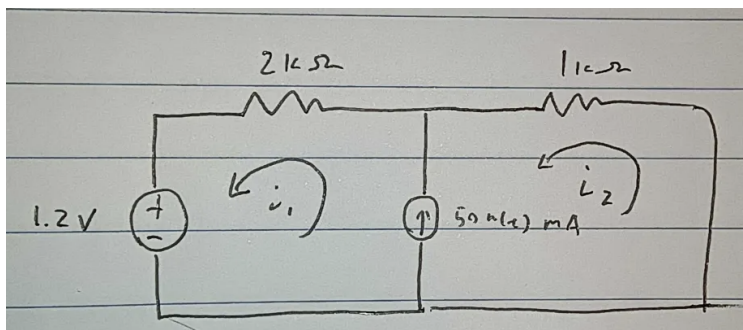
$$\tau = \frac{L_{\text{eq}}}{R_{\text{eq}}} = \frac{[10 \times 10^{-3} \text{ H}]}{[0.75 \times 10^3 \Omega]} = 13.33 \times 10^{-6} \text{ s}$$

Now find the initial condition and steady state conditions for the current through the inductor after turning the sources back on.

Initially ($t = 0$), the inductor behaves like a short circuit under the 1.2 V source, but the second source is still off. Thus, the current through the inductor is the same as the current through both resistors in series:

$$V_s = iR_{\text{eq}} \implies i = i_L(0) = \frac{1.2}{3 \times 10^3} = 0.4 \text{ mA}$$

At steady state ($t \rightarrow \infty$), the inductor behaves like a short circuit, thus the current through the inductor is equal to the current through the $1 \text{ k}\Omega$ resistor. Quick mesh analysis gives us:



Supermesh eqn:

$$\begin{aligned} i_1 - i_2 &= 50 \times 10^{-3} \\ -[1 \text{ k}\Omega]i_2 - [2 \text{ k}\Omega]i_1 - [1.2 \text{ V}] &= 0 \end{aligned}$$

Solving for i_1 and i_2 :

$$i_1 = 0.01626 \text{ A}, \quad i_2 = -0.0337 \text{ A}$$

Thus, following the sign convention for i_L :

$$i_L(\infty) = -i_2 = 0.0337 \text{ A}$$

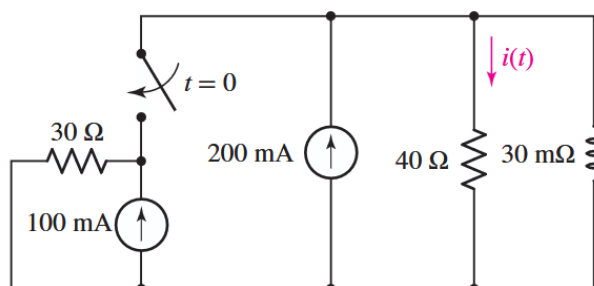
Finally, using the textbook's formula for driven RL circuits:

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]e^{-t/\tau} = 0.0337 - 0.0333e^{-t/(13.33 \times 10^{-6})} [\text{A}]$$

(b)

$$\begin{aligned} i_L(10 \times 10^{-6}) &= 17.99 \times 10^{-3} \text{ A} \\ i_L(10 \times 20^{-6}) &= 26.30 \times 10^{-3} \text{ A} \\ i_L(10 \times 50^{-6}) &= 32.95 \times 10^{-3} \text{ A} \end{aligned}$$

- 8.67** 67. Obtain an expression for $i(t)$ as labeled in the circuit diagram of Fig. 8.91, and determine the power being dissipated in the 40Ω resistor at $t = 2.5 \text{ ms}$.



■ **FIGURE 8.91**

Following the textbook method again, but first find the initial condition (recall that while current is continuous, voltage across an inductor can change instantaneously). Thus the initial voltage is equal to the voltage across the resistors in parallel:

$$R_{\text{eq}} = \frac{30 \cdot 40}{30 + 40} = 17.14 \Omega$$

At $t = 0$, $i_L(0^-) = i_L(0^+) = 200 \text{ mA}$, so the voltage across the resistors is given by the remaining current:

$$i_{\text{source}} = i_{\text{resistors}} + i_L(0^+) \implies i_{\text{resistors}} = 100 \text{ mA}$$

$$v_L(0^+) = [17.14 \Omega] \cdot [100 \times 10^{-3} \text{ A}] = 1.714 \text{ V}$$

After $t = 0$, the switch is flipped and the time constant is given by:

$$\tau = \frac{L_{\text{eq}}}{R_{\text{eq}}}$$

Turning off the sources to find the equivalent inductance and resistance:

$$R_{\text{eq}} = 17.14 \Omega$$

$$L_{\text{eq}} = 30 \times 10^{-3} \text{ H}$$

$$\tau = \frac{30 \times 10^{-3} \text{ H}}{17.14 \Omega} = 1.75 \times 10^{-3} \text{ s}$$

At steady state ($t \rightarrow \infty$), the inductor shorts the entire circuit:

$$v_L(\infty) = 0 \text{ V}$$

Thus,

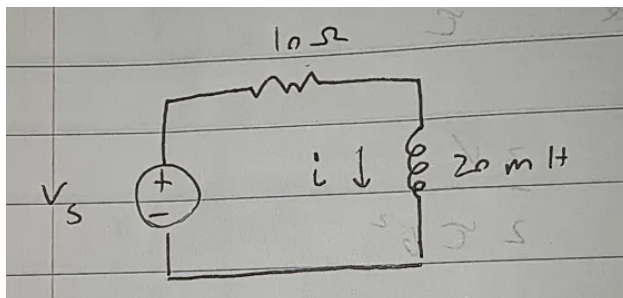
$$v_L(t) = v_L(\infty) + [v_L(0) - v_L(\infty)]e^{-t/\tau} = 1.714e^{-t/(1.75 \times 10^{-3})} [\text{V}]$$

Now, the voltage across the 40Ω resistor is the same as the voltage across the inductor, so the power dissipated by the 40Ω resistor is given by:

$$v_L(2.5 \times 10^{-3}) = 0.4107 \text{ V}$$

$$P = \frac{V^2}{R} = \frac{[0.4107 \text{ V}]^2}{[40 \Omega]} = \boxed{0.004218 \text{ W}}$$

- 8.70** 70. A dc motor can be modeled as a series RL circuit (though with the addition of a back emf voltage, which we will ignore in this problem), where rotational speed is proportional to the current flowing in the circuit. A motor with $R = 10 \Omega$ and $L = 20 \text{ mH}$ has a source voltage of 2.5 V that is suddenly increased to 5 V . How long would it take for the motor to reach 95 percent of full speed?



Given:

- $\omega = ki$
- $v_s = 2.5 + 2.5u(t) \text{ V}$

The question asks for *when* the motor reaches 0.95 of full speed. Since motor speed is proportional to the current through the circuit, we simply have to find the time to reach 0.95 of the steady state current.

Textbook method for solving driven RL circuits again. The time constant is given by:

$$\tau = \frac{L}{R} = \frac{[0.02 \text{ H}]}{[10 \Omega]} = 0.002 \text{ s}$$

The initial condition is given by the current through the inductor at $t = 0^+$ since the current is continuous:

$$i_L(0^-) = \frac{[2.5 \text{ V}]}{[10 \Omega]} = 0.25 \text{ A}$$

$$i_L(0^+) = i_L(0^-) = 0.25 \text{ A}$$

At steady state ($t \rightarrow \infty$):

$$i_L(\infty) = \frac{[2.5 + 2.5 \text{ V}]}{[10 \Omega]} = 0.5 \text{ A}$$

Thus, using the textbook's formula for driven RL circuits:

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]e^{-t/\tau} = 0.5 - 0.25e^{-t/0.002} [\text{A}]$$

Thus, 0.95 of the steady state current is given by:

$$0.95 \cdot 0.5 = 0.475 \text{ A}$$

Solve for t :

$$0.475 = 0.5 - 0.25e^{-t/0.002} \implies t = \boxed{4.605 \times 10^{-3} \text{ s}}$$